Chapter 5

Bolted Riveted and Welded Connections

5.1 Question Bank UNIT - 5 AS

Quiz Questions

1. In a lap joint, at least ..... bolts should be provided in a line.
2. Use of lap joints is not recommended because .... are distributed unevenly
3. A rivet is specified as a 20mm rivet. What does it mean?
4. Riveting is preferred in joining .... materials.
5. The necessary diameter (d) for thickness of plates (t) in a riveted joint is......
6. Tearing of the plate between the holes is taken when ......
7. In a lap joint, if the plates are connected to each other when the joint is made with only one row of rivets then it is called ....-riveted lap joint.
8. In butt joint when one strap is used the thickness varies between .... to .... (T is the thickness of plate to be connected).
9. Rivets in group subjected to direct loads share load ....
10. Edge distance is measured ...... to direction of stress, while end distance is measured ..... to direction of stress
11. ....joint have high corrosion resistance
12. .... type joint is not fillet welded
13. Thickness of cover plate when only one cover plate is used, t1= ....t
14. The centre to centre distance between two consecutive rivets in a row, is called.......
15. A lap joint is always in .....shear.
16. The efficiency of a riveted joint is defined as the ratio of the .... to the strength of the un-riveted or solid plate.
17. The strength of a joint may be de

    ned as the maximum ...., which it can transmit, without causing it to fail

18. A joint may fail due to tearing of the plate at an edge. This can be avoided by keeping the margin, m = ...

19. When the rivets in the varoius rows are opposite to each other then the joint is said to be ..... riveted

20. Of the three types of structural joints, the fail safe based joint is ....

Main Questions

1. What are the various permanent and detachable fastenings? Give a complete list with the different types of each category.

2. Show by neat sketches the various ways in which a riveted joint and welded joint may fail.

3. Give examples of eccentrically loaded welded joints. How are they analysed?

4. Draw the plan and elevation for the following riveted joints. 1. Single riveted lap joint 2. Double riveted lap joint having zig zag riveting 3. Double riveted lap joint having chain riveting

5. Find the efficiency of the following riveted joints : 1. Single riveted lap joint of 6 mm plates with 20 mm diameter rivets having a pitch of 50 mm. 2. Double riveted lap joint of 6 mm plates with 20 mm diameter rivets having a pitch of 65 mm. Assume Permissible tensile stress in plate = 120 MPa Permissible shearing stress in rivets = 90 MPa Permissible crushing stress in rivets = 180 MPa

6. A double riveted lap joint is made between 15 mm thick plates. The rivet diameter and pitch are 25 mm and 75 mm respectively. If the ultimate stresses are 400 MPa in tension, 320 MPa in shear and 640 MPa in crushing, a

   nd the minimum force per pitch which will rupture the joint. If the above joint is subjected to a load such that the factor of safety is 4, fi

   nd out the actual stresses developed in the plates and the rivets.

7. What do you mean by 'Design of a riveted joint '? While designing a riveted joint how will you fi

   nd different quantities?

8. A single riveted double cover butt joint is used to connect two plates of 15mm thick. the rivets are 26mm in diameter and are provided at a pitch of 10cm. The allowable stresses in tension shear and crushing are 130 Mpa 75 Mpa and 150 Mpa. Find the safe load per pitch length of the joint and efficiency of the joint.

9. Find the efficiency of the following riveted joints :

   (a) Single riveted lap joint of 6 mm plates with 20 mm diameter rivets having a pitch of 50 mm.

(b) Double riveted lap joint of 6 mm plates with 20 mm diameter rivets having a pitch of 65 mm.

Assume Permissible tensile stress in plate = 120 MPa Permissible shearing stress in rivets = 90 MPa Permissible crushing stress in rivets = 180 MPa

10. A $90 \times 50 \times 8$ mm angle carrying a load pf 125kN is to be connected to a gusset plate by welding. If the size of the weld is 6mm and max allowable shear stress in the weld is 102.5N/mm^2, find the lengths of the weld at the top and bottom. The distances between the neutral axis and the bottom and the top edges of the angle sections are 28.7mm and 61.3mm respectively.